THE ALBANESE METHOD OF PROJECTION

ANGELINE PENG

1. Introduction

One of the main questions in classical algebraic geometry is the resolution of singularities. Recall that there is an equivalence of categories between function fields over k (finitely generated transcendence degree 1 field extensions of k) and smooth projective curves over k. While obtaining the function field associated to the curve is straightforward, it is not necessarily true that the curve obtained from a function field by looking at its relations defines a smooth curve. What is true is that there is always a (unique) smooth curve birational to this potentially singular curve, and since they are birational, they have the same function field. In general, a smooth projective curve that is birational to a singular one is called its **resolution**.

In this paper, we shall approach this problem using the Albanese method of projection [1], a classical approach to this problem that solves the question of resolution for curves with degree that is not too high. The algorithm involves iteratively picking singular points as the center for a stereographic projection of the curve onto a hyperplane. In this case, each of these projection maps will be birational and the algorithm shall terminate in a smooth curve living in a smaller-dimensional space.

In the first section, we recall some definitions necessary for the proof of the theorem. The classical treatment of curves often assumes that the curves are smooth. In this section, we shall clarify the necessary results in our singular case. The second section contains the proof of the theorem. The main reference for the proof is [3]. The third section will be an application of this algorithm to singular elliptic curves. In particular, we shall use this to show the classical result that all curves defined by singular Weierstrass equations are birational to the projective line $\mathbb{P}^1$.

2. Definitions

Let k be an algebraically closed field and C/k an irreducible and reduced projective curve. We shall write $k[C]$ to denote the ring of regular functions on C and $k(C)$ to denote the function field of C.

2.1. Singularities and Multiplicities.

We say a point is smooth if it locally looks like a linear subspace.

Definition 2.1. Let C be a projective curve. We may view it as the vanishing locus $Z(f_1, \ldots, f_{n-1}) \subseteq \mathbb{P}^n$ of the homogeneous polynomial $f_i \in k[x_0, \ldots, x_n]$. A

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point $p \in C$ is **smooth** if the Jacobian at that point

$$J(p) = \begin{bmatrix} \frac{\partial f_1}{\partial x_0} & \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_0} & \frac{\partial f_2}{\partial x_1} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & & \ddots & \vdots \\ \frac{\partial f_{n-1}}{\partial x_0} & \cdots & & \frac{\partial f_{n-1}}{\partial x_n} \end{bmatrix}$$

is full rank. Otherwise, we say that p is a **singularity**.

We introduce the notion of multiplicity to measure how much the point fails to be smooth. Pick a point $p \in C$. If p is a smooth point, then the local ring at the point $\mathscr{O}_{C,p}$ is a discrete valuation ring. For an element $f \in k(C)^\times$, we may then define the valuation of f at p to be $v_p(f)$ where v_p is the valuation on $\mathscr{O}_{C,p}$ extended to its field of fraction $\mathrm{Frac}(\mathscr{O}_{C,p}) = k(C)$.

We do not have this luxury when p is not smooth. Instead, we hope to emulate this by noting that in the smooth case, we may write $f = t^l u$ where $t \in \mathfrak{m}_p$ and $u \in \mathscr{O}_{C,p}$ some unit. Then, $v_p(f) = l = \dim_k \mathscr{O}_{C,p}/f\mathscr{O}_{C,p}$. We shall use this as the basis for the definition for the multiplicity of a function.

Definition 2.2. The **multiplicity** of $f \in k[C]$ at p is

$$e_p(f) = \dim_k \mathscr{O}_{C,p}/f\mathscr{O}_{C,p}.$$

For $f \in k(C)^\times$, we may write it as $f = \frac{g}{h}$, for $g, h \in k[C]^\times$. Then, we define

$$e_p(f) = e_p(g) - e_p(h).$$

Now we shall define the multiplicity of a point on the curve. This quantity measures how bad a singularity is by looking at the minimum vanishing order of functions that vanish at the point. Intuitively, if a point is a singularity, it has multiple branches passing through it, and so the order of vanishing of any function at that point must be higher.

Definition 2.3. The **multiplicity** of a point $p \in C$ is given by

$$\mathrm{mult}_p(C) = \min_{f \in \mathfrak{m}_p} (e_p(f))$$

We note that when p is a smooth point, $\mathscr{O}_{C,p}$ is a DVR and we may take f to be the uniformizer of $\mathfrak{m}_p$. Therefore, $\mathrm{mult}_p(C) = \dim_k \mathscr{O}_{C,p}/\mathfrak{m}_p = \dim_k k = 1$.

2.2. Degrees of Curves and Maps.

Definition 2.4. Let C be a projective curve embedded in $\mathbb{P}^n$. Let $H = Z(f) \subseteq \mathbb{P}^n$ be a hyperplane that does not completely contain C. The **degree** of C is the intersection number of C and H. In this case, this can be computed as

$$\deg C = \sum_{p \in C \cap H} \dim_k \mathscr{O}_{C,p}/f\mathscr{O}_{C,p}$$

Remark 2.5. We could take the above sum over all points of C as at the points $p \notin C \cap H$, $f \in \mathscr{O}_{C,p}^\times$ and the resulting dimension is 0. Moreover, using the notation of Definition 2.3,

$$\deg C = \sum_{p \in C} e_p(f).$$

Next we define the degree of a rational map. As we'll see in Lemma 2.8, the degree of a surjective map is related to the number of preimages of each point.

Definition 2.6. Given a rational map of curves $\varphi : C \to C'$, it induces a map on function fields $\varphi^* : k(C') \to k(C)$ by precomposition. We define the **degree of** φ to be

$$\deg \varphi = [k(C) : \varphi^* k(C')]$$

Corollary 2.7. *A rational map is birational if and only if it has degree 1.*

Fact 2.8. *Let $\varphi : C \to C'$ be a surjective rational map between curves. Let $p \in C'$ be a smooth point such that $\varphi^{-1}(p)$ is a finite collection of smooth points. Then, for all but finitely many such p,*

$$|\varphi^{-1}(p)| = \deg \varphi$$

The above fact is usually stated for C' a smooth curve, but its proof only requires that the point p and its preimages are smooth. One reference for this proof is Theorem 1 of section III.2 of [4].

2.3. **Stereographic Projection and the Blowup.**

Definition 2.9. Let $p \in \mathbb{P}^n$ be a point and let $H \subseteq \mathbb{P}^n$ be a hyperplane that does not contain p. The **stereographic projection** with center p to H is the rational map $\pi : \mathbb{P}^n \dashrightarrow H \cong \mathbb{P}^{n-1}$ defined by mapping each point $q \in \mathbb{P}^n$ to $l_{pq} \cap H$, where the line passing through p and q intersects with the hyperplane.

Under a change of basis, we may assume that $H = (x_0 = 0)$ and $p = [1 : 0 : \cdots : 0]$. Then,

$$\pi_p : [x_0 : \cdots : x_n] \mapsto [x_1 : \cdots : x_n]$$

If p happens to be a smooth point, then we may take l_{pp} to be the tangent line to C at p. However, when p is not smooth, there are multiple tangents, and so the map is not well-defined. To remedy this, we take the blowup of C at p. The blowup separates all the tangent lines passing through p into distinct points, and we may then extend our projection map from C to the blowup of C.

Definition 2.10. The **blowup of** $\mathbb{P}^n$ at a point $[1 : 0 : \cdots : 0]$ is the subvariety $B_0\mathbb{P}^n$ of $\mathbb{P}^n \times \mathbb{P}^{n-1}$ defined by the equations $x_i y_j = x_j y_i$ for all $1 \leq i, j \leq n$ where $x_0, x_1, \ldots, x_n$ are the coordinates of $\mathbb{P}^n$ and $y_1, \ldots, y_n$ are the coordinates of $\mathbb{P}^{n-1}$. Define the map $\varphi_1 : B_0\mathbb{P}^n \to \mathbb{P}^n$ to be the projection map from the product $\mathbb{P}^n \times \mathbb{P}^{n-1}$ to the first factor and $\varphi_2 : B_0\mathbb{P}^n \to \mathbb{P}^{n-1}$ to be the projection onto the second factor.

We may similarly define $B_p\mathbb{P}^n$ for some $p \in \mathbb{P}^n$ by making a change of coordinates such that $p \rightsquigarrow [1 : 0 : \cdots : 0]$. Suppose we have a projective variety $C \subseteq \mathbb{P}^n$ and a point $p \in C$. The **blowup of** C at p is given by the projective closure $B_pC = \overline{\varphi^{-1}(C \setminus \{p\})}$.

For a more detailed discussion of blow ups, see the associated section in 1.4. of [2]. In particular, $C \setminus \{p\}$ is isomorphic to $C' \setminus \varphi_1^{-1}(p)$, and $\varphi_1^{-1}(p)$ consists of all the tangent directions of C through p. Thus, we extend π_p to B_pC by mapping $\varphi_1^{-1}(p) \mapsto \varphi_2(\varphi_1^{-1}(p)) \in \mathbb{P}^{n-1}$. This becomes a morphism of curves with image

$$\pi_p(B_pC) = \pi_p(C \setminus \{p\}) \cup \varphi_2(\varphi_1^{-1}(p)) = \overline{\pi_p(C)}$$

3. The Albanese Method

3.1. **The Algorithm.** Given an irreducible, reduced projective curve $C \subseteq \mathbb{P}^n$ that spans all of $\mathbb{P}^n$, we define the Albanese method as follows: let $C_0 = C$. For $i = 0, 1, \ldots,$ pick $p \in C_i$ a singular point. Let π_p be the stereographic projection with center p to some hyperplane. Define $C_{i+1} = \overline{\pi_p(C_i)} \subseteq \mathbb{P}^{n-(i+1)}$. Terminate when C_i is smooth.

Theorem 3.1. *If $\deg C < 2n$, then each projection is a birational map and this procedure eventually terminates with a smooth curve.*

3.2. **Proof of Theorem.** We begin with a few lemmas regarding how the degree of the curve behaves under the projection map.

Lemma 3.2. *Let $C \subseteq \mathbb{P}^n$ be an irreducible and reduced curve that is not contained in any hyperplane. Then, if we pick n points $p_1, \ldots, p_n \in C$, $\deg C \geq \sum_i \operatorname{mult}_{p_i} C$. In particular, if all these points are smooth, then $\deg C \geq n$.*

Proof. Given our n points $p_1, \ldots, p_n$, we may find a hyperplane $H = Z(f)$ passing through these points. Since C is not contained in a hyperplane,

$$\deg C = \sum_{p \in C} \dim_k \mathscr{O}_{C,p}/f\mathscr{O}_{C,p} \geq \sum_{i=1}^{n} \dim_k \mathscr{O}_{C,p_i}/f\mathscr{O}_{C,p_i} \geq \sum_{i=1}^{n} \operatorname{mult}_{p_i}(C)$$

with the last inequality being true as $f \in \mathfrak{m}_{p_i}$ for each i. If these points are smooth, then $\operatorname{mult}_{p_i}(C) = 1$, and so $\deg C \geq n$. □

Lemma 3.3. *Let C_0 be an irreducible reduced curve and $p \in C$. Let π_p be the projection map with base point p. Let $C_1 = \overline{\pi_p(C_0)}$. Then,*

$$\deg C_1 \cdot [k(C_0) : k(C_1)] = \deg C_0 - \operatorname{mult}_p(C_0)$$

Proof. Let $H \subseteq \mathbb{P}^{n-1}$ be a hyperplane satisfying that is does not intersect C_1 at any singular points. We know that H intersects C_1 at $\deg C_1$ points. Now consider the intersection $\pi^{-1}(H) \cap C_0$. Note that $\pi^{-1}(H)$ is a hyperplane as well, and it intersects C_0 at the preimages of the points in $H \cap C_1$. For every point in the intersection, there are $[k(C_0) : k(C_1)]$ points in the preimage by Fact 2.8 (if we pick H appropriately. Note that these are Zariski closed conditions and so we may choose such a general hyperplane H). Additionally, $\pi^{-1}(H)$ also intersects C_0 at p with multiplicity $\operatorname{mult}_p(C)$. Thus,

$$\deg C_0 = \deg C_1 \cdot [k(C_0) : k(C_1)] + \operatorname{mult}_p(C_0)$$

Rearranging yields us the statement of the lemma. □

Proof of theorem. We proceed via induction. For $n = 1$, we notice by Lemma 3.2 that $1 \leq \deg C < 2$, so $\deg C = 1$. Furthermore, $\deg C = 1 \geq \operatorname{mult}_p(C)$ for all $p \in C$, which implies that all points of C are smooth.

Now for the inductive case. First, we note that C_{i+1} is still irreducible and reduced as rational maps preserve these properties. Next, it still spans all of $\mathbb{P}^{n-1}$. If C_{i+1} does not, then it is contained within some hyperplane H. That would then mean that $\pi^{-1}(H) \supseteq C_i$, a contradiction.

Now we show that each projection is indeed a birational map. Suppose for the sake of contradiction that π_p is not birational. Then, $[k(C_0) : k(C_1)] \geq 2$.

Furthermore, recall that since p is singular, $\mathrm{mult}_p(C_i) \geq 2$. Thus, applying Lemma 3.3 we obtain

$$\deg C_{i+1} \leq \frac{\deg C_i - \mathrm{mult}_p(C_i)}{2} < \frac{2(n-i)-2}{2} = n - (i+1)$$

But by Lemma 3.2, we know that $\deg C_{i+1} \geq n - (i+1)$. We conclude that π_p must be birational.

By Lemma 3.3, $\deg C_{i+1} = \deg C_i - \mathrm{mult}_p(C_i) < 2(n-i) - 2 = 2(n-(i+1))$. We may then apply the inductive hypothesis to this curve of smaller degree $C_{i+1} \subseteq \mathbb{P}^{n-(i+1)}$. □

4. Application to Singular Elliptic Curves

For this section, assume that char $k \neq 2, 3$. It is perhaps bad practice to say singular elliptic curves, as elliptic curves are often by definition nonsingular. What we mean by a singular elliptic curve is a plane curve given as the vanishing locus of the dehomogenized equation $y^2 = x^3 + ax + b$ where $a, b \in k$ such that the discriminant of the cubic in x vanishes. In other words, $\Delta = 4a^3 + 27b^2 = 0$.

Let us verify that this curve is indeed singular. The Jacobian is

$$\begin{bmatrix} 2y & -3x^2 - a \end{bmatrix}$$

The singularities are then given by points $(r, 0)$ where r is a root of $x^3 + ax + b$. Note that since the discriminant is zero, r is at least a double root and so satisfies $3r^2 + a = 0$. Notably, since any singular point must be a multiple root of this cubic, there is exactly one singularity. We may then assume that the singularity is at the origin. This leaves us with two main classes of singular elliptic curves: nodal elliptic curves of the form $y^2 = (x-a)^2(x-b)$ and cuspidal elliptic curves of the form $y^2 = (x-a)^3$.

Theorem 4.1. *Every singular elliptic curve is birational to* $\mathbb{P}^1$.

Proof. Let C be a singular elliptic curve. One can check that it is irreducible and reduced. We notice that $\deg C = 3 < 2 \cdot 2$. Thus, we may apply the Albanese algorithm to C. It terminates in one step. The projection map from origin to the hyperplane given by $z = 0$ is given explicitly as

$$\pi_0 : [x : y : z] \mapsto [x : y]$$

Furthermore, this map is surjective as fixing values for x and y yields a cubic in z. Over an algebraically closed field k, we may always find a root. □

Remark 4.2. The classical birational map between $k(C)$ and $k(t)$ arises from this map. After restricting to an affine chart, this morphism is given by $(x, y) \mapsto \frac{y}{x}$ with inverse $t \mapsto (t^2 + a, t(t^2 + a))$ in the nodal case and $t \mapsto (t^2, t^3)$ in the cuspidal case.

References

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Massachusetts Institute of Technology
Email address: pengs14@mit.edu